

Mehrdad Kashefi

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Education	Western University , London, ON, Canada <i>Neuroscience Ph.D. Candidate</i> Preliminary Thesis Title: “Neural Basis of Sequential Action” Advisors: Dr. Andrew Pruszynski, Dr. Jörn Diedrichsen	2020 - now
	Iran University of Science and Technology (IUST) , Tehran, Iran <i>M.S. Electrical Engineering - Bioengineering</i>	2017 - 2020
	Lorestan University , Lorestan, Iran <i>B.S. Electrical Engineering - Electronics</i>	2013 - 2017
Awards	COSYNE Presenters Travel Grant–USD \$500	2025
	Neural Control of Movement (NCM) travel award	2024
	International Ontario Graduate Scholarship–CAD \$15,000	2023
	Best Oral Presentation at Neuroscience Research Day, Western University	2023
	Neuroscience Program Travel Award, Western University–CAD \$500	2022
	Graduated First in Class (Highest GPA), IUST	2020
Publications		
Preprints	[2] Kashefi M. , Michaels J., Kersten R., Lau J., Diedrichsen J., Pruszynski J.A. “Compositional neural dynamics during reaching”, bioRxiv 10.1101/2025.09.04.674069	
	[1] Michaels J., Kashefi M. , Zheng J., Codol O., Weiler J., Kersten R., Lau, J., Gribble P., Diedrichsen J., Pruszynski J.A. “Sensory expectations shape neural population dynamics in motor circuits”, bioRxiv 10.1101/2024.12.22.629295	
Published	[8] Kashefi M. , Diedrichsen J. [†] , Pruszynski J.A. [†] “Motor sequence learning involves better prediction of the next action and optimization of movement trajectories”, Journal of Neuroscience.e0299252025 († := Co-senior authors)	
	[7] Shahbazi M., Ariani G., Kashefi M. , Pruszynski J.A., Diedrichsen J., “Neural correlates of online action preparation”, Journal of Neuroscience.e1880232024	
	[6] Kashefi M. , Reschechtko S., Ariani G., Shahbazi M., Alice Tan, Diedrichsen J., Pruszynski J.A. “Future movement plans interact in sequential arm movements”, eLife.94485.1,2024	
	[5] Chung B., Zia M., Thomas K., Michaels J., Jacob A., Pack A., Williams M., Nagapudi K., Teng L., Arrambide E., Ouellette L., Oey N., Gibbs R., Anschutz P., Lu J., Wu Y., Kashefi M. , Oya T., Kersten R., Mosberger A., O’Connell S., Wang R., Marques H, Rita P. Mendes A., Lenschow C., Kondakath G., Kim J., Olson W.,Quinn K., Perkins P., Gatto G., Thanawalla A., Coltman S., Kim T., Smith T., Binder-Markey B., Zaback M., Thompson C., Giszter S., Person A., Goulding M., Azim E., Thakor N., O’Connor D., Trimmer B., Lima S., Carey M., Pandarinath C., Costa R., Pruszynski A., Bakir M., Sober S. “Myomatrix arrays for high-definition muscle recording”, eLife.88551.1, 2023	
	[4] Codol O., Michaels J. A, Kashefi M. , Pruszynski J.A., Gribble P. “MotorNet: a Python toolbox for controlling differentiable biomechanical effectors with artificial neural networks”, eLife.88591.1, 2023	

- [3] Codol O., **Kashefi M.**, Forgaard C.J., Galea J.M., Pruszynski J.A., Gribble P. “Sensorimotor feedback loops are selectively sensitive to reward”, eLife 12:e81325, 2023
- [2] Ahmadi A.*, **Kashefi M.***, Shahrokhi H, Nazari M.A., “Computer aided diagnosis system using deep convolutional neural networks for ADHD subtypes”, Biomedical Signal Processing and Control, 2021 (* := equal contribution)
- [1] **Kashefi M.**, Daliri M.R., “A stack LSTM structure for decoding continuous force from local field potential signal of primary motor cortex (M1)”, BMC bioinformatics, 2021

Conference

Talks

Kashefi M., Michaels J. A, Diedrichsen J., Pruszynski J.A., “Motor cortical dynamics during reaching connect posture-specific attractors”, Neural Control of Movement (NCM), Panama City, Panama, 2025

Kashefi M., Diedrichsen J., Pruszynski J.A., “A compositional solution for sequence learning”, Neural Control of Movement (NCM), Dubrovnik, Croatia, 2024 (Online)

Kashefi M., Diedrichsen J., Pruszynski J.A., “Is motor learning merely anticipation?”, Neuroscience Research Day (NRD), London, Canada, 2023

Posters

Kashefi M., Michaels J. A, Diedrichsen J., Pruszynski J.A., “Motor cortical dynamics during reaching connect posture-specific attractors”, Cosyne, Montreal, Canada, 2025

Kashefi M., Diedrichsen J., Pruszynski J.A., “Is there more to sequence learning than better anticipation?”, Neural Control of Movement (NCM), Victoria, Canada, 2023

Kashefi M., Ariani G., Diedrichsen J., Pruszynski J.A., “Planning multiple future actions in sequential reaching”, Neural Control of Movement (NCM), Dublin, Ireland, 2022

Invited talks

Kashefi M., “Motor cortical dynamics during reaching connect posture-specific attractors”, Chinese institute for brain research, July 2025 (Online)

Kashefi M., “Future movement plans interact in sequential arm movements”, Motor control Group, West Virginia University, December 2024 (Online)

Teaching

Mentorship

Armin Panjehpour (PhD, Western University, 2023-)

Alice Tan (BSc, Western University, 2022-2023)

Workshop

Introduction to Machine Learning, Brainhack Western 2021

TA

Physiology and Pharmacology Laboratory, Western University 2021

Computational Neuroscience, IUST 2019-2020

Electronic Circuits, IUST 2018

Outreach

Mentorship Committee, Society of Neuroscience Graduate Students, Western 2022
Mentoring first-year graduate students in the neuroscience program.

Thames Valley Science and Engineering Fair, London, ON, CA 2021-2023

Judge for Grade 4-12 Science Engineering Fair

Python Camp, IUST 2019

Introductory python course for middle school students

Executive Committee Member, ICBME , IUST
26th International Iranian Conference on Biomedical Engineering (ICBME)

2019

Referees

Dr. Andrew Pruszynski

Associate Professor, Canada Research Chair in Sensorimotor Neuroscience
Western University
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Dr. Jörn Diedrichsen

Professor, Western Research Chair for Motor Control and Computational Neuroscience
Western University
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Dr. Jonathan Michaels

Assistant Professor
York University
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